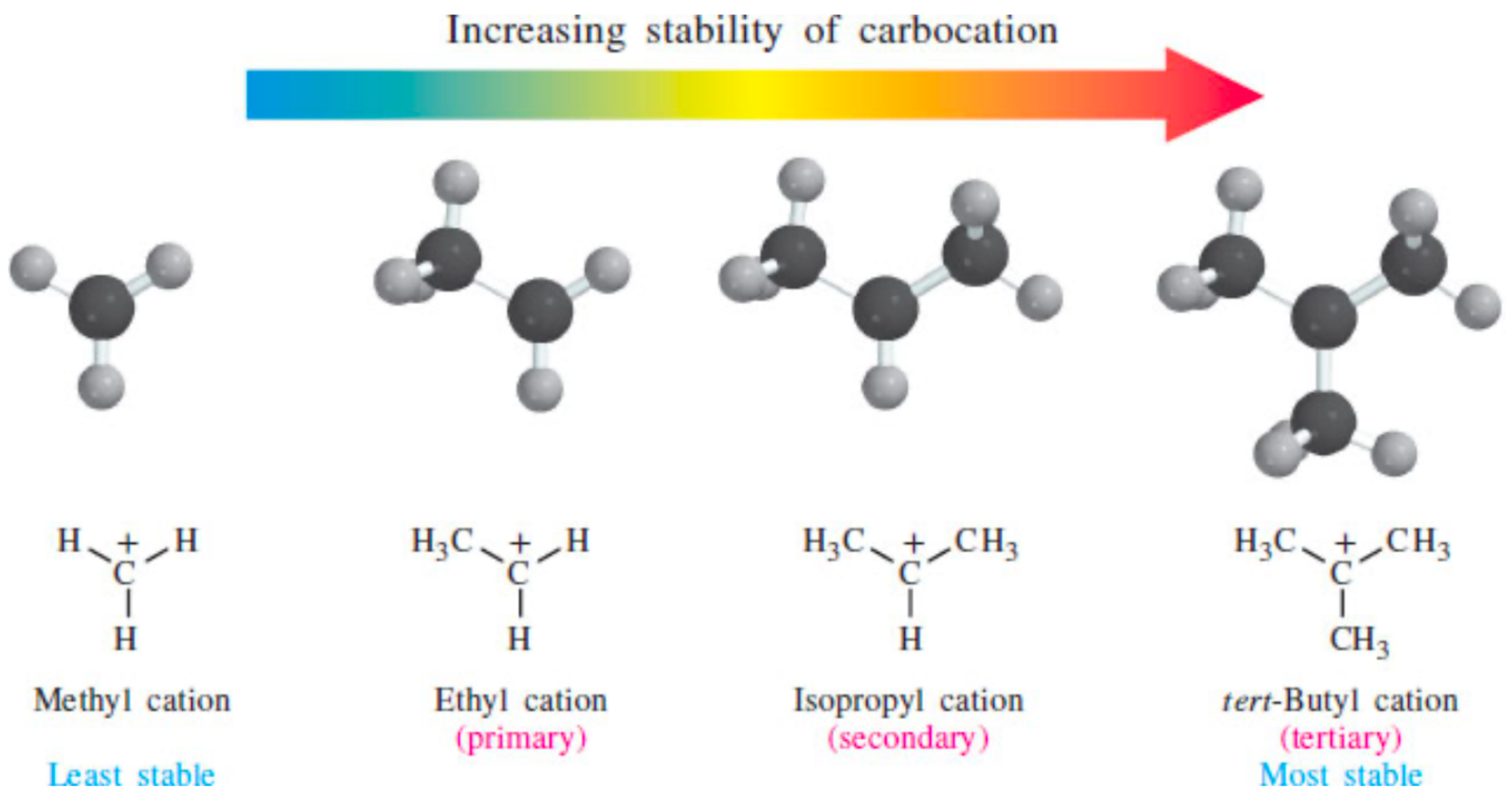
QX4001 – Molecules to Materials

Rotation 3 – Question Set 2

1) Why does the stability of carbocations follow the trend: tertiary > (more stable than) secondary > primary > methyl?

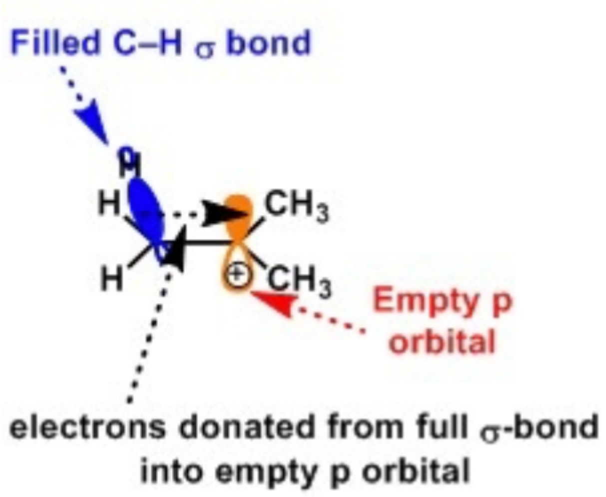
Consider both hyperconjugation, electronic (separately).

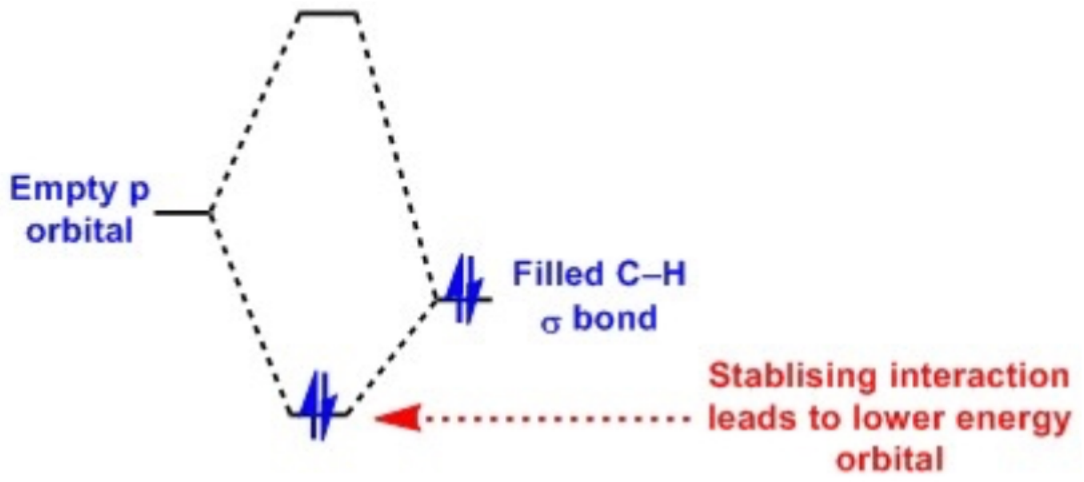
1) Why does the stability of carbocations follow the trend: tertiary > (more stable than) secondary > primary > methyl?
Consider both hyperconjugation, electronic (separately). 各自的



1) Why does the stability of carbocations follow the trend: tertiary > (more stable than) secondary > primary > methyl?
Consider both hyperconjugation, electronic (separately).

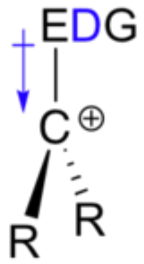
Hyperconjugation: Carbocations, are stabilized by substituents such as alkyl groups that can donate electrons to the unfilled orbital by hyperconjugation, where the unpaired electron, plus electrons in σ bonds that are α to the radical site, are delocalized. So more substituents, results in an increased stability.



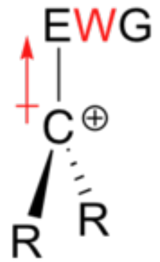


1) Why does the stability of carbocations follow the trend: tertiary > (more stable than) secondary > primary > methyl?
Consider both hyperconjugation, electronic (separately).

What will stabilize a positive charge? **An electron donating group!**



Electron Donating Group: stabilizes a carbocation



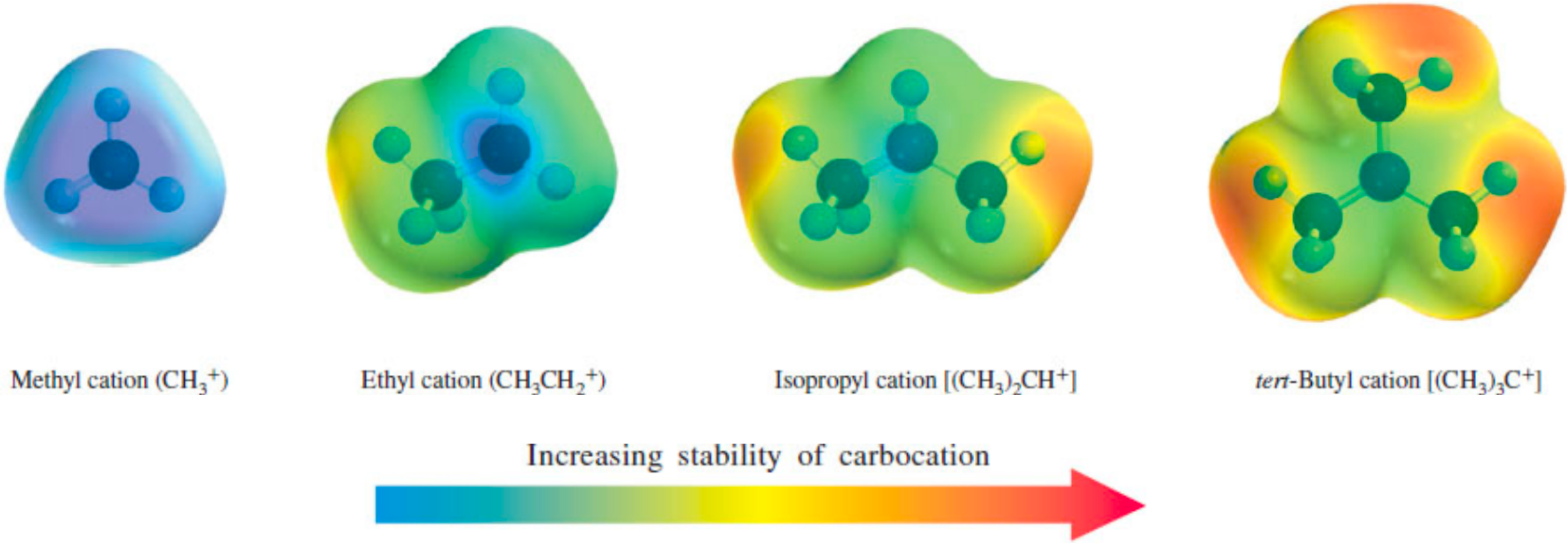
Electron Withdrawing Group: *de*stabilizes a carbocation

A positively charged species such as a carbocation is very electron-poor, and thus anything which donates electron density to the electron poor carbon center will help to stabilize it.
Alkyl groups are weak electron donating groups (**Positive Inductive effect +*I***), and thus stabilize nearby carbocations.

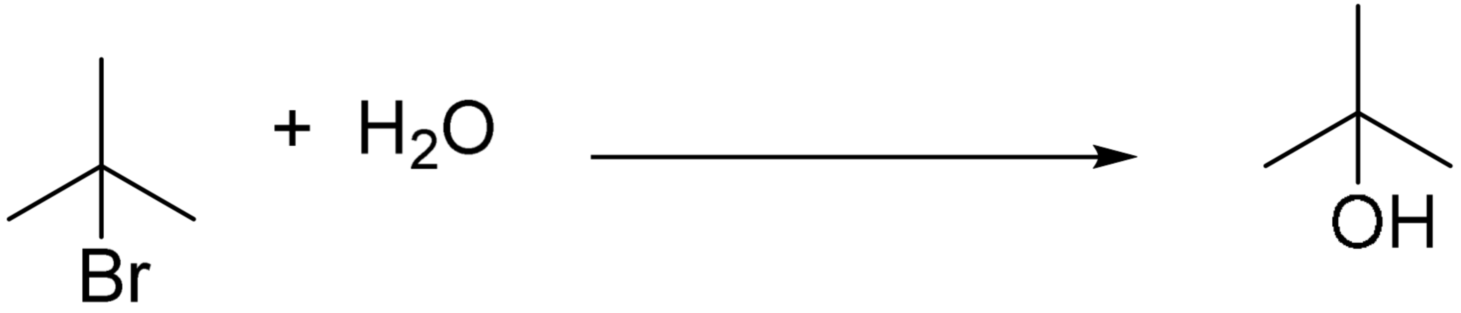
Recall that an inductive effect is an electron-donating-or-withdrawing effect of a substituent that is transmitted by the polarization of σ bonds. Electrons in a C–C bond are more polarizable than those in a C–H bond, so replacing hydrogens by alkyl groups reduces the net charge on the positively charged carbon. Alkyl groups are electron-releasing substituents with respect to their inductive effect. The more alkyl groups that are directly attached to the positively charged carbon, the more stable the carbocation.

1) Why does the stability of carbocations follow the trend: tertiary > (more stable than) secondary > primary > methyl?
Consider both hyperconjugation, electronic (separately).

Inductive effect (by polarization of σ bonds)
Hyperconjugation (by delocalization of electrons in σ bonds)



2) For the following reaction, identify the type of mechanism and draw it, and discuss what type of solvent you would use (discuss the properties of the solvent required, and propose specific solvents)

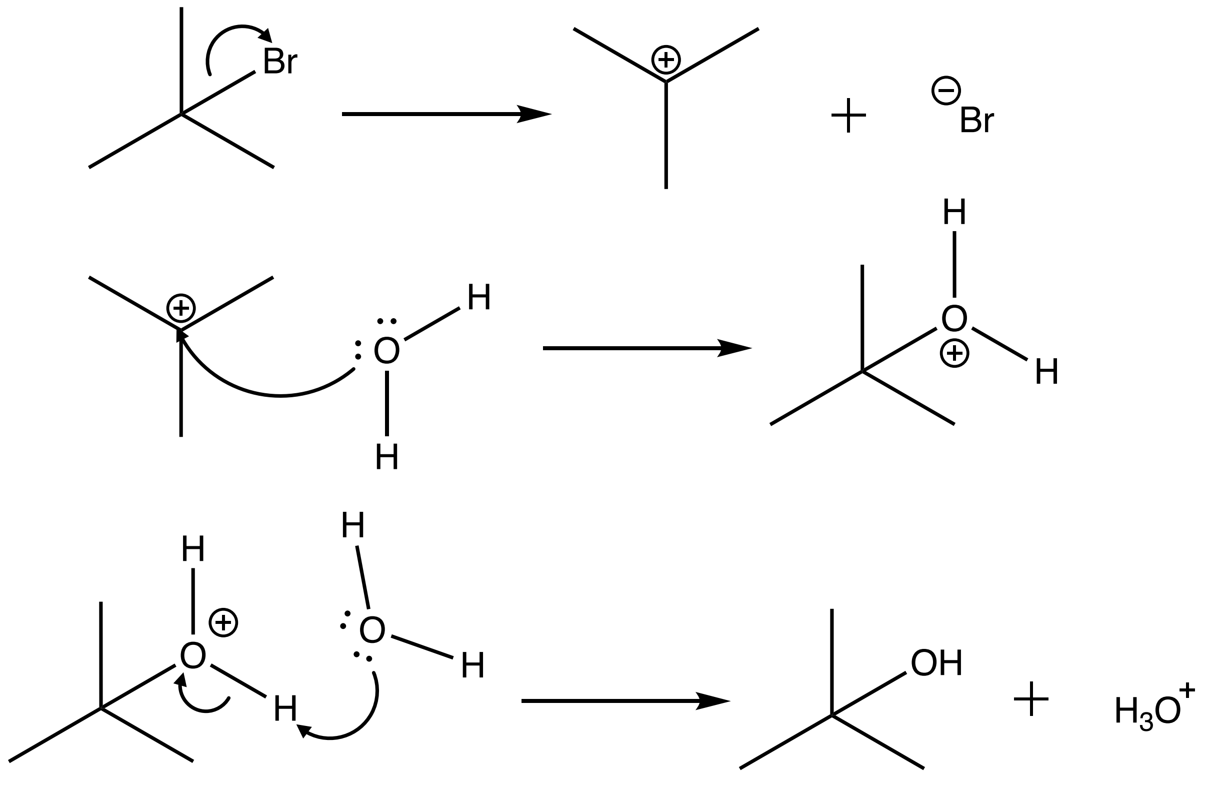


2) For the following reaction, identify the type of mechanism and draw it, and discuss what type of solvent you would use (discuss the properties of the solvent required, and propose specific solvents)

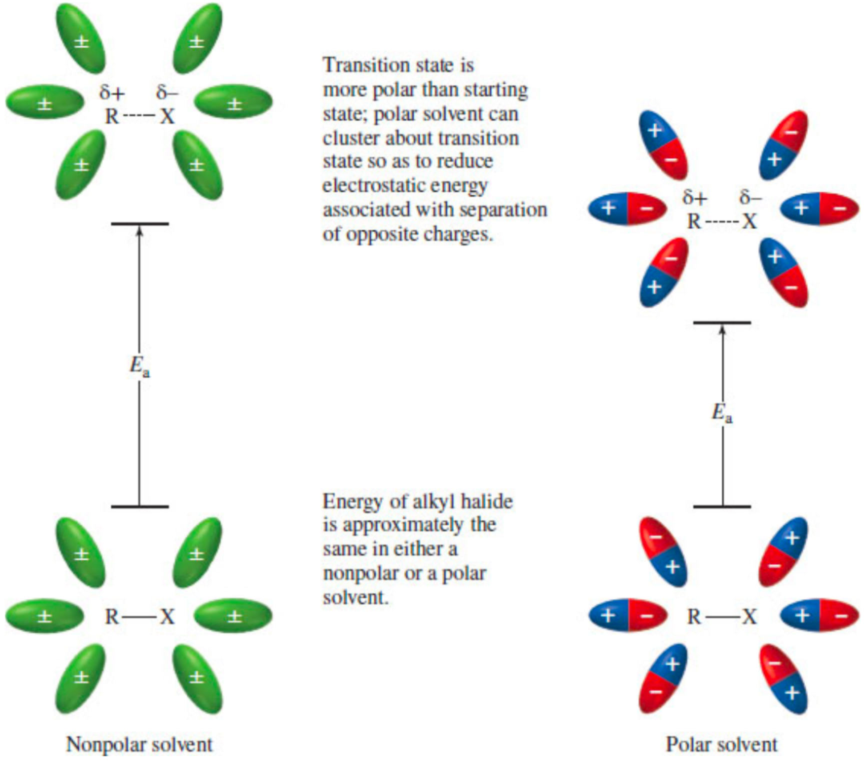
Step 1

Step 2

Step 3



2) For the following reaction, identify the type of mechanism and draw it, and discuss what type of solvent you would use (discuss the properties of the solvent required, and propose specific solvents)

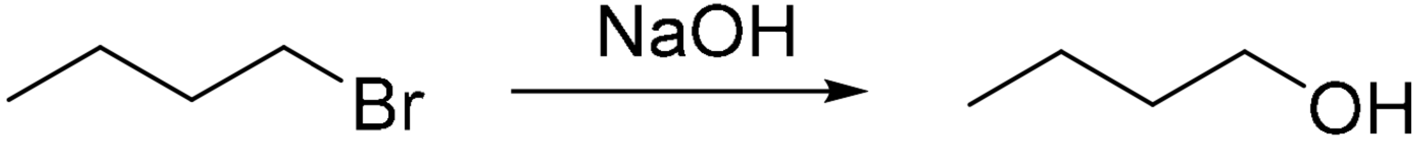


Solvent chosen: Aqueous Ethanol

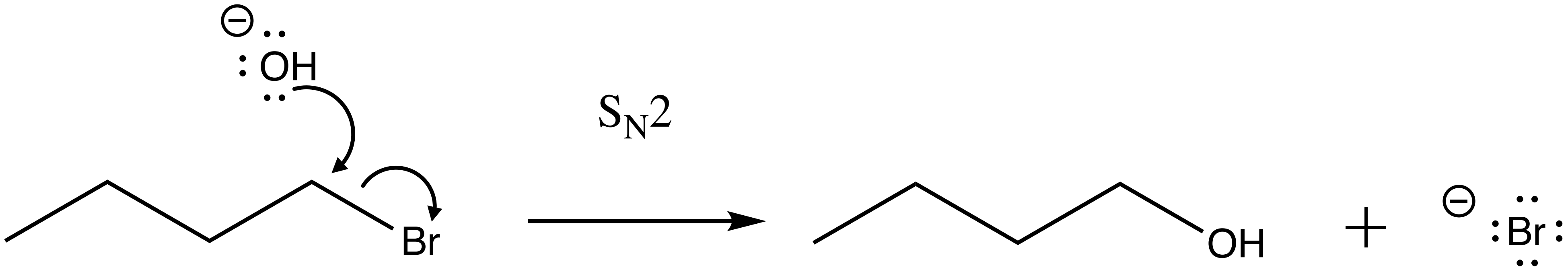
Polar and nonpolar solvents are similar in their interaction with the starting alkyl halide, but differ markedly in how they **stabilize the transition state.** A solvent with a low dielectric constant has little effect on the energy of the transition state, whereas one with a high dielectric constant (**Polar solvent)** stabilizes the charge- separated transition state, lowers the activation energy, and increases the rate of the reaction.

Since the hydrogen atom in a polar protic solvent is highly positively charged, it can interact with the anionic nucleophile which would negatively affect an S_N2, but it does not affect an S_N1 reaction because the nucleophile is not a part of the rate-determining step. Polar protic solvents actually speed up the rate of the unimolecular substitution reaction because the large dipole moment of the solvent helps to stabilize the transition state. The highly positive and highly negative parts interact with the substrate to lower the energy of the transition state. Since the carbocation is unstable, anything that can stabilize this even a little will speed up the reaction.

3) For the following reaction, identify the type of mechanism and draw it, and discuss what type of solvent you would use (discuss the properties of the solvent required, and propose specific solvents)



3) For the following reaction, identify the type of mechanism and draw it, and discuss what type of solvent you would use (discuss the properties of the solvent required, and propose specific solvents)

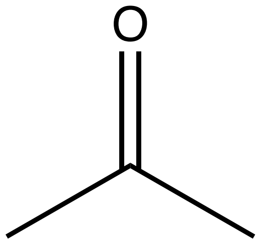


A polar APROTIC solvent is preferred. Aprotic solvents, like protic solvents, are polar but, because they lack a positively polarized hydrogen, they do not form hydrogen bonds with the anionic nucleophile (which would decrease nucleophilicity).

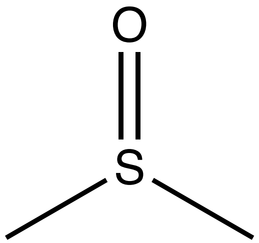
For this reason we chose to use Acetone (other good answers would be DMSO, DMF)

3) For the following reaction, identify the type of mechanism and draw it, and discuss what type of solvent you would use (discuss the properties of the solvent required, and propose specific solvents)

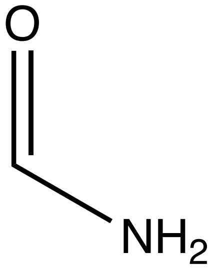
Polar Aprotic solvents:



acetone



DMSO
dimethyl sulfoxide



DMF
N,N-dimethylformamide

4) Consider the following reaction, which could proceed by either SN1 or S_N2.

Discuss the effect on k_{SN1}/k_{SN2} when you change the following conditions:

a) X: Cl, I
b) Y: H_2O, NaOH
c) Solvent: Acetone, Aqueous Ethanol

4) Consider the following reaction, which could proceed by either SN1 or S_N2.

Discuss the effect on k_{SN1}/k_{SN2} when you change the following conditions:

a) X: Cl, I

$$\left(\frac{k_{S_N1}}{k_{S_N2}}\right) I \quad > \quad \left(\frac{k_{S_N1}}{k_{S_N2}}\right) Cl$$

Good leaving groups are weak bases. They're happy and stable on their own. The weaker the base, the **better** the **leaving group**.

I$^-$ is a better leaving group than ***Cl***$^-$, so S_N1 will be easier for ***I***$^-$.

4) Consider the following reaction, which could proceed by either SN1 or S_N2.

Discuss the effect on k_{SN1}/k_{SN2} when you change the following conditions:

b) Y: H_2O, NaOH

$$\left(\frac{k_{S_N1}}{k_{S_N2}}\right)H_2O \;>\; \left(\frac{k_{S_N1}}{k_{S_N2}}\right)OH^-$$

For S_N1 the nucleophilicity of the nucleophile does not really matter that much. The effect will be more pronounced for the S_N2 reaction.
S_N2 is favoured for strong nucleophiles, usually bearing a negative charge.
So in this case S_N2 will go faster with ***OH***$^-$.

4) Consider the following reaction, which could proceed by either SN1 or S_N2.

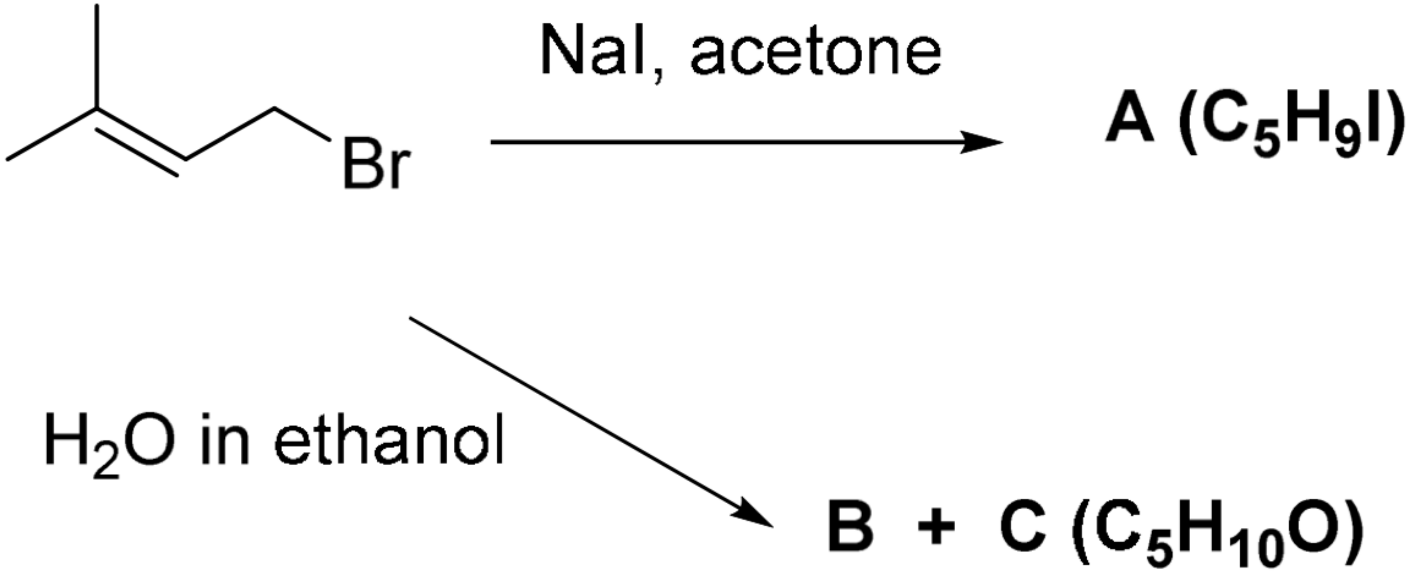
Discuss the effect on k_{SN1}/k_{SN2} when you change the following conditions:

c) Solvent: Acetone, Aqueous Ethanol

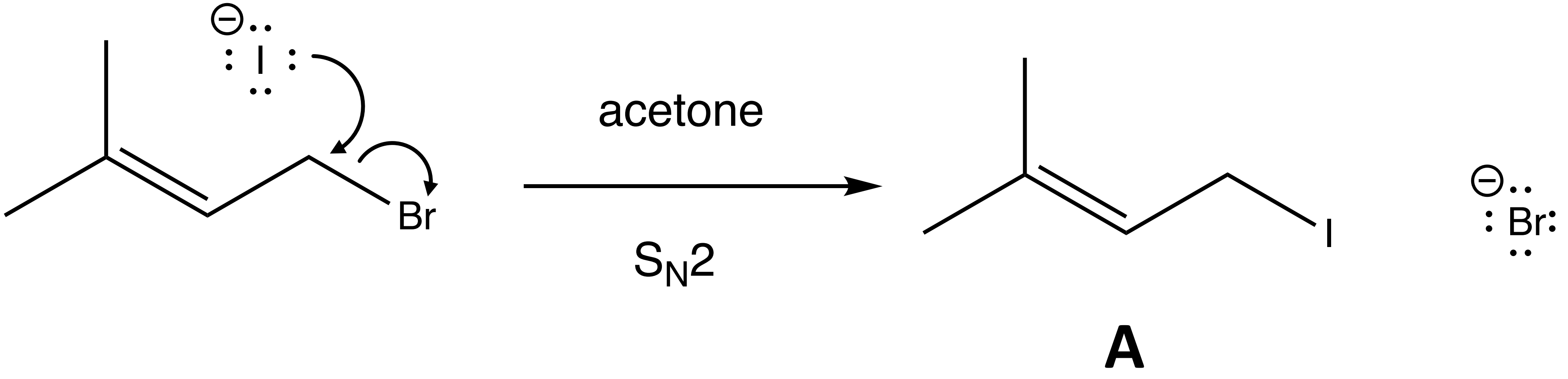
$$\left(\frac{k_{S_N1}}{k_{S_N2}}\right) acetone \quad < \quad \left(\frac{k_{S_N1}}{k_{S_N2}}\right) aqueous\ ethanol$$

S_N1 is favoured in **polar protic solvents** they stabilize the transition state and carbocation intermediate however they also make nucleophiles less nucleophilic by solvating them, **S_N2** is favoured in **polar aprotic solvents,** they can dissolve the nucleophile (as opposed to apolar solvents) and don't solvate as much as protic solvents..

5) Propose structures for A, B and C, and draw mechanisms for the reactions below. B and C are structural isomers. B is the major product of the second reaction.



5) Propose structures for A, B and C, and draw mechanisms for the reactions below. B and C are structural isomers. B is the major product of the second reaction.



5) Propose structures for A, B and C, and draw mechanisms for the reactions below. B and C are structural isomers. B is the major product of the second reaction.

S_N1 H_2O in ethanol